

Homework 21

Please read Section 6.3 of Durrett. The two examples at the end of the section (about investing) are interesting—maybe Francisco and Patrick will like them! The formulas for expected value, variance, and standard deviation of the mean-value random variables for an iid sequence will be useful for answering the first question below. In the question, though, I want you to justify your reasoning using properties of expected value and variance and not just to quote those formulas.

In our remaining classes, we will study the Central Limit Theorem, which tells us that the random variables $\overline{X_n}$ for n large are close to normally distributed. You have probably heard about normally distributed random variables in your statistics classes and maybe in your calculus classes.

The standard normal distribution is the continuous probability distribution with density function f given by the formula $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$; possible values of a random variable with this distribution range from $-\infty$ to ∞ . In Section 6.4, Durrett uses multivariable calculus (polar coordinates) to show that f is a density function, explaining how π gets into probability theory. (You do not need to follow this calculation!) Durrett also shows that if X has the standard normal distribution, then $E[X] = 0$ and $\text{Var}(X) = 1$.

Please also read Section 6.4 to learn about the normally distributed random variables.

1. In Homework 20, you thought about rolling a die repeatedly. In the die-rolling situation, let Y_i be the indicator random variable for “you get a 6 on the i th roll.”
 - a. Calculate $E[Y_1]$ and $\text{Var}(Y_1)$.

The variable Y_1 has the Bernoulli(1/6) distribution. We thus have $E[Y_1] = 1/6$ and $\text{Var}(Y_1) = \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)$.

- b. Consider the random variable $\overline{Y_{20}} = \frac{1}{20}(Y_1 + Y_2 + \cdots + Y_{20})$. This random variable is the proportion of 6s in the first 20 rolls. Use the “formulaire” for expected value and variance to calculate $E[\overline{Y_{20}}]$, $\text{Var}(\overline{Y_{20}})$, and $\sigma(\overline{Y_{20}})$. Show enough of your work to justify your conclusions.

By the scaling and additivity properties of expected value, we have the following:

$$E[\overline{Y_{20}}] = \frac{1}{20}(E[Y_1] + \cdots + E[Y_{20}]) = \frac{1}{20}\left(20 \cdot \frac{1}{6}\right) = \frac{1}{6}.$$

By the scaling property of variance and the independence of the Y_i , we have the following:

$$\text{Var}(\overline{Y_{20}}) = \frac{1}{20^2}(\text{Var}(Y_1) + \cdots + \text{Var}(Y_{20})) = \frac{1}{20^2}\left(20 \cdot \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)\right) = \frac{1}{20}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right) = \frac{5}{720} = \frac{1}{144}.$$

$$\sigma(\overline{Y_{20}}) = \sqrt{\frac{1}{144}} = \frac{1}{12}.$$

- c. Repeat the calculations of expected value, variance, and standard deviation for $\overline{Y}_n = \frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)$, where n is a positive integer.

By the same reasoning, we have $E[\overline{Y}_n] = 1/6$, $\text{Var}(\overline{Y}_n) = \frac{1}{n}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)$, and $\sigma(\overline{Y}_n) = \frac{1}{\sqrt{n}}\left(\frac{\sqrt{5}}{6}\right)$.

- d. What are the limiting values for $E[\overline{Y}_n]$, $\text{Var}(\overline{Y}_n)$, and $\sigma(\overline{Y}_n)$? What do these limiting values tell you about the proportion of 6s in a large number of rolls of a die?

The limiting value of $E[\overline{Y}_n]$ is $1/6$: the long run proportion of 6s tends to be around $1/6$.

The limiting value of $\text{Var}(\overline{Y}_n)$ and $\sigma(\overline{Y}_n)$ is 0. In the long run, the proportion is very likely to be very close to $1/6$.

These facts together are essentially the weak law of large numbers for repeating the chance experiment of rolling a die and recording whether the roll is a 6.

2. The following question is a version of Durrett's Exercise 5 from Chapter 6.

- a. 10 people get on an elevator on the first floor of a seven-story building. Each gets off at one of the six higher floors chosen at random. What is the probability that at least one person gets off on floor 2?

Let A be the event that at least one person gets off at floor 2. Then $P(A) = 1 - P(A^c)$. The event A^c is that no people get off on floor 2. We assume that the 10 events "person i gets off at floor 2" are independent and that each of these events has probability $1/6$. We thus have $P(A^c) = (5/6)^{10}$ and $P(A) = 1 - (5/6)^{10} \approx 0.8385$.

- b. Let N be the random variable counting the number of stops the elevator makes. Use the method of indicators to calculate $E[N]$. Explain your method, showing how to write N as a sum of indicator random variables. You must name the indicators and the events that they indicate. (Durrett's answer from the back of the book: 5.03.)

Let A_i for $i = 2, \dots, 7$ be the event that at least one person gets off at floor i . By the calculation above, we have $P(A_i) = 1 - \left(\frac{5}{6}\right)^{10}$. Let X_i be the indicator for A_i . Then $N = X_2 + \cdots + X_7$. Therefore, $E[N] = E[X_2] + \cdots + E[X_7] = 6\left(1 - \left(\frac{5}{6}\right)^{10}\right) \approx 5.0310$.

In Section 6.4, Durrett tells us that to calculate probabilities for events related to a normally distributed random variable, we need to use a table. These days, we can also use calculators and online tools such as Desmos, but we will learn the old-fashioned methods for reasons that I will explain in class.

3. Suppose that $X \sim \text{normal}(0, 1)$, i.e. X is a random variable with the standard normal distribution. Use the table in the back of the book ("Normal Table" in the table of contents) to calculate the following probabilities:
- a. $P(X \leq 1.23)$

According to the normal table, this probability is 0.8907.

- b. $P(X > 1.5)$

We calculate this probability as $1 - P(X \leq 1.5) \approx 1 - 0.9332 = 0.0668$.

c. $P(-0.68 \leq X \leq 0.68)$

We calculate this probability as $P(X \leq 0.68) - P(X < -0.68)$. The symmetry of the normal distribution tells us that $P(X < -0.68) = P(X > 0.68) = 1 - P(X \leq 0.68)$. Therefore, the desired probability is $2P(X \leq 0.68) - 1 \approx 2(0.7517) - 1 = 0.5034$.

As a rule of thumb, I remember that for normally distributed data, about 50% of the data fall within 2/3 standard deviation of the mean.

d. $P(-1.96 \leq X \leq 1.96)$

By the same reasoning as above, $P(-1.96 \leq X \leq 1.96) = 2P(X \leq 1.96) - 1 \approx 2(0.9750) - 1 = 0.9500$.

This calculation justifies the 95% rule: for normally distributed data, about 95% of the data fall within 2 (or 1.96) standard deviations of the mean.

A key fact from Section 6.4 is that if $Y \sim \text{normal}(\mu, \sigma^2)$, then $\frac{Y-\mu}{\sigma} \sim \text{normal}(0, 1)$. Durrett mentions this fact just before Example 6.20. That example is a calculation based on this fact. You may know $\frac{Y-\mu}{\sigma}$ as the z-score; it measures the values of the random variable as deviations from the mean value μ in units of the standard deviation σ . You will use the method Durrett uses in Example 6.20 to solve the following problem.

4. The heights of US women (measured in centimeters) roughly follow the normal(162, 6.5²) distribution. Let Y be the height of a randomly chosen US woman, which we model as $Y \sim \text{normal}(162, 6.5^2)$. Use the method described in Example 6.20 to calculate $P(Y \leq 175)$.

We first calculate the z-score for 175. It is $\frac{175-162}{6.5} = 2$.

Therefore, $P(Y \leq 175) = P\left(\frac{Y-162}{6.5} \leq 2\right) \approx 0.9772$.